

The Internet — past, present and future

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This paper charts the transformation of the Internet from its military and academic origins to its current status as a global network for multimedia information publishing and retrieval. It explains the basic technologies that have enabled this transformation and introduces the advances that will underpin its future development. These include security and electronic commerce, network computing and three-dimensional worlds, and the development of an integrated service platform for the delivery of real-time audio and video services.

1. Introduction and background

For some years now many people have had a vision of a global information superhighway which would support a host of interactive, multimedia applications in fields such as education, medicine, entertainment and commerce. The superhighway would be global, ubiquitous, intelligent, and have sufficient bandwidth to provide true multimedia services. The Internet is the nearest that we have come to reaching that goal on a global scale. The Internet can already claim to be global, and although there is no ubiquitous broadband access, the number of users is growing dramatically. Its potential to support electronic commerce and enable businesses to reach a global market has been widely recognised. Developments in security and electronic payment technologies are overcoming concerns over security, and are encouraging the introduction of a wide range of commerce applications and services. Search engines enable users to find information on a vast range of subjects, and the Internet's potential to support education and training is already being realised with the launch of the first educational services, such as BT's CampusWorld [1].

Although designed as a data network, the Internet can also support audio and video traffic and new network protocols are being developed which will provide for efficient network usage and enable quality of service guarantees. Combined with the dramatic improvement in the processing power of personal computers, the potential for a global multimedia communications platform is clearly there. However, the widespread usage of audio and video applications on the Internet would require significantly increased bandwidth in the Internet backbone (or better interconnectivity with high-bandwidth circuit-switched networks via gateways) as well as increased access

bandwidths. Indeed, the Internet is already suffering growing pains, and network providers have struggled to keep up with the demand.

Despite current limitations, the Internet has already had a profound impact on the world's communications and information technology sectors, where it is stimulating an unprecedented rate of technological development. Internet technologies are also used to advantage in corporate networks, or intranets. The need to supply global interconnectivity for these corporate networks has provided the impetus for more capacity to be furnished by the major global carriers on the global Internet backbone.

Although the Internet came to the world's notice in the 1990s, it was in fact born, a product of the Cold War, in the 1960s. It began life as the ARPANET, an experiment in reliable, decentralised networking by the US Department of Defense (DoD), linking the DoD with military contractors and universities conducting military-funded research. Over the next decade the network grew as an increasing number of North American universities and research bodies joined. In 1973 the first European organisations were connected. By the early 1980s the development of Unix was leading to further rapid growth in networked computing, particularly in universities. A second network, MILNET, took over the military traffic, while European (EUNET) and Japanese Unix networks became established. In the UK, JANET (Joint Academic Network) was formed. A further important development took place in 1986 when the US National Science Foundation established NSFNET to link five US university supercomputers. During the late 1980s, as regional networks in the US and networks in more and more

countries were connected, what was by now called the Internet became truly global. It was no longer solely an academic network — Internet access was available from the home. In 1990 the ARPANET was shut down and NSFNET took over the administration of the Internet. In 1991 the restriction on commercial use of the Internet was lifted and by 1994 several large commercial networks had grown within it — these now dominate the global network in terms of traffic carried.

The capabilities of the Internet have also grown steadily, with key developments including the introduction of electronic mail, and the file transfer protocol (FTP) in the 1970s, the domain name system and TCP (a reliable transport protocol) in the early 1980s. Then in 1989, Tim Berners-Lee of CERN proposed the World Wide Web (WWW) which became the key application of the 1990s by transforming the Internet into a global, multimedia information service. It was the WWW which attracted a much broader spectrum of users and led to the explosive growth of recent years, and today's estimated 30 to 50 million Internet users world-wide. As the capabilities of the Internet have increased, so its usage has changed. For many years the Internet was used for communication and data sharing mainly by academics and government researchers. It is now used widely from the home and the office to access a vast pool of information published on the WWW. A wide range of on-line services have evolved, including shopping, 3-D worlds and audio and video applications including telephony and videoconferencing.

The basic capabilities of the Internet and WWW for communication and publishing are described in section 3 of this paper. Later sections introduce the more advanced developments that are underpinning the development of electronic commerce and the transformation of the Internet into an integrated services platform which also supports real-time audio and video services.

2. Who controls the Internet?

It can easily be argued that no-one controls the Internet and that it will continue to grow organically in response to both commercial and technological pressures. Nevertheless, there are several bodies that oversee the evolution of the technical standards on which the Internet is based. The ultimate authority for the development of the Internet rests with the Internet Society [2], which was formed in January 1992 to provide long-term support for the Internet standards process carried out by the Internet Engineering Task Force (IETF). The Society's individual and organisational members share common goals of maintaining the long-term viability of the Internet and its continued evolution and global scaling. The Society is incorporated as a not-for-profit organisation and is governed by a Board of Trustees elected by its membership around the world. In June 1992, the Internet Architecture Board (previously called the Internet

Activities Board) proposed associating its activities with the Internet Society. The IAB [3] provides oversight of the architecture for the protocols used by the Internet and the processes used to create them.

The complex and detailed work of developing and standardising the Internet protocol suite is undertaken by the IETF. The IETF [4] began in 1986 to enable technical co-ordination by the US Department of Defense contractors working on ARPANET and related networks. The IETF is now an extremely large open international community of network designers and researchers representing a wide range of organisations and interests including universities, equipment and software vendors, and network operators. The technical work of the IETF is done in its working groups, which are organised to cover major topics such as routing, security and network management. The IETF meets three times a year, but most of the work is done off-line via mailing lists. At the 37th IETF meeting in December 1996, there were approximately 2000 attendees from around the world. The IETF has been rapidly expanding the capabilities of the Internet protocol (IP) suite over the last two years, targeting in particular the issues of real-time traffic and quality of service, as well as developing new intelligent networking and service paradigms such as multicast and mobile IP. There is also a strong push to internationalise the Internet, and in particular to enable it to support the full ISO 10646 character set which contains characters for most languages.

The operational management of the Internet standards process is handled by the Internet Engineering Steering Group (IESG) under the auspices of the Internet Society.

The development of the WWW transformed the Internet into a global, multimedia information service and led to further dramatic growth in its user population. It also led to the formation of another 'standards' organisation — the World Wide Web Consortium (W3C) [5] was founded in 1994 to develop common standards for the evolution of the Web. It is an industry consortium, jointly hosted by the Massachusetts Institute of Technology Laboratory for Computer Science (MIT/LCS) in the United States, the Institut National de Recherche en Informatique et en Automatique (INRIA) in Europe, and the Keio University Shonan Fujisawa Campus in Asia. To date, the W3C has more than 150 member organisations.

The work of the W3C includes hypertext markup language (HTML), the integration of object technology into the Web, and developments aimed at making the Web faster through improvements to HTTP (hypertext transfer protocol). The W3C is addressing concerns over pornography on the Internet through the development of its platform for content selection (PICS). This protocol enables content labelling and rating for the Web and makes it possible for people to control the type of content they (or

their children) see. To stimulate electronic commerce, the W3C initiated the Joint Electronic Payment Initiative (JEPI), a project that addressed the need for a standardised way of negotiating payment methods between buyers and sellers on the WWW.

3. Basic components of the Internet

To allow the Internet to develop rapidly, it has evolved around many different component technologies each playing a specific role. This section introduces the main components of the Internet (and tries to explain any associated jargon). A fuller description can be found in the book by Krol [6].

3.1 Internet protocol

The foundation of the Internet is the Internet protocol. IP is a network protocol for routing units of data, called packets, across the network from one host computer to another (Fig 1 illustrates the interconnection of hosts, routers and networks). IP is very simple because it focuses specifically on routing. Consequently it makes no guarantees about delivering packets. Packets may be lost or corrupted, and a sequence of packets sent between two hosts may take different routes and may arrive in a different order. These issues are left for other protocols to handle.

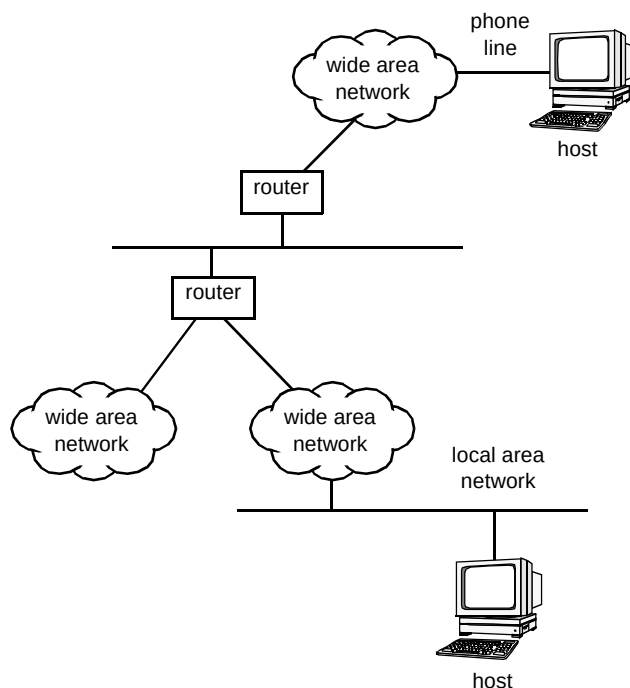


Fig 1 Host, routers and networks.

IP packets are sent from one router to another towards their destination. Each router maintains routing tables

which say on which interface a packet should be sent in order to progress towards its destination. On arrival at a router a packet is stored in a queue. When it reaches the front of the queue the router reads the destination address in the packet header, looks it up in the routing table and then transmits the packet via the appropriate interface.

If the network is congested, then the input queues at routers can become long, increasing the transit delay. Under heavy congestion the queues may become full and packets are discarded.

IP was originally designed for use in ARPANET which had to be robust and flexible enough to withstand wartime attacks. Routing tables are regularly updated to take account of network failures or changes to the configuration of the network.

IP has established itself as the *de facto* internetworking standard for two main reasons. Firstly, the lack of delivery-guarantees make it simple and efficient (not all applications require these guarantees). Secondly, its ability to adapt to network failures and configuration changes has made IP particularly robust and has enabled it cope with the exponential growth that has occurred on the Internet in recent years. Networks of many sizes and type can be easily connected.

3.2 Transport protocols

In order to provide error checking, error correction or recovery, transport protocols are used together with IP. The transport frame (including data and transport header) is placed inside the Internet packet, i.e. it forms the data part of an IP packet.

There are two main transport protocols in use on the Internet — UDP and TCP.

- User datagram protocol (UDP)

UDP is a 'fire-and-forget' protocol intended for sending messages without guarantee of arrival and without notifying the sender of successful or failed delivery. It is very simple; in addition to the data carried within the packet, the packet only provides information about the application which is using it and a checksum to tell whether the data has been corrupted in transit.

- Transmission control protocol (TCP)

In contrast to UDP, TCP is designed to handle all manner of network failures. If packets are lost or corrupted then TCP arranges for them to be resent. If

packets arrive out of order then TCP will reorder them. If packets are repeatedly lost then the TCP source assumes that the network is congested and adapts its transmission rate accordingly.

3.3 Ports

At any one time a host computer may be engaged in many ‘conversations’ with other hosts, and a number of different applications may be involved. Port numbers are used to identify for which application and for which conversation a packet is intended. Each UDP or TCP packet carries two port numbers — one port number identifies the server application, the other port number being selected by the client to distinguish the conversation from any other. Well-known ports (used with common applications like Telnet and FTP) are defined in the Assigned Numbers RFC. An RFC is a Request for Comments and is part of the standardisation process used by the IETF.

3.4 Telnet

Telnet is the most basic of all Internet functions. It simply allows a user to access a server remotely — a function known as remote terminal access. It works by carrying ASCII text typed by the user to the remote server and returning the output from the application on the remote server to the user.

3.5 File transfer protocol (FTP)

FTP was the earliest information retrieval service on the Internet and brought the shareware industry to life. It provides a basic service for the reliable transfer of files from one machine to another. FTP allows the user to establish a control connection between their client and the server. They can then use this connection to navigate through the servers directory structure and request the transmission of files. In order to transmit files a separate data connection is set up.

3.6 Domain name service (DNS)

The domain name service provides name/address translation for all objects on the Internet. Every computer (i.e. every host and every router) on the Internet has a name. However, its fully distinguished name depends on a hierarchy of names and is written by concatenating the hierarchy of names using ‘.’ as a separator, e.g. transend.labs.bt.com

In this name the top level domain is ‘com’ and BT owns the sub-domain or name-space ‘bt.’ Any owner of a name space must run a DNS server and give its address to the owner of the DNS server at the next level up. Applications such as FTP, SMTP, Telnet, and WWW send a request to the local DNS server which responds either with the answer or with the address of a DNS server which can answer. If it responds with the answer then this may have come from its

own tables or from another DNS server (which it approached on the application’s behalf).

3.7 E-mail

Electronic mail or e-mail is the electronic equivalent of the traditional postal service. E-mail is sent from a mail client (a program which runs on the user’s machine) to the destination mailbox using the simple mail transfer protocol (SMTP). Other protocols such as POP3 or IMAP4 are used for checking and retrieving mail from a mailbox. SMTP routes from the mail agent to the destination mailbox via a number of mail handlers which behave just like post office sorting offices, i.e. they sort the mail and pass it on to the next mail handler.

An SMTP mail address is based on Internet domain names — specifically they take the form user@domain., e.g. rudkins@boat.bt.com. In order to route the message to an e-mail address of this form, the mail agent depends on DNS to translate the domain part of the e-mail address (the part after ‘@’) into an IP address.

A blessing and a curse of e-mail is that it is as easy to send a message to a list of people as it is to a single person. Used with care e-mail lists are a great way of informing a group of people. Used poorly, or without respect for the recipients, e-mail from lists may be little more than junk mail.

3.8 Bulletin boards

An alternative to e-mail lists is bulletin boards or newsgroups. These are the electronic equivalent of traditional notice-boards. They allow users to post messages for viewing by the wider community. Anyone can reply to the message and the resulting sequences of messages arising from an initial posting is grouped into a discussion thread.

Both e-mail lists and bulletin boards may be used to support group working. The main difference is in the level of user participation required. New e-mail messages are sent to each user on a mailing list automatically, whereas the user must choose to go to the bulletin board to read new postings.

3.9 World Wide Web

The application that has generated the recent explosion in the use of the Internet is of course the World Wide Web. The WWW makes publishing, display and retrieval of all kinds of media as easy as clicking a mouse. This ease of use, combined with the availability of free software and the existence of a large networked community made the WWW irresistible. The World Wide Web now accounts for most of the traffic on the Internet.

Documents on the WWW are usually written in hypertext markup language which describes the layout of the page and the components such as the text and images that it contains. HTML pages are stored at servers known as web servers and the information is accessed using a client application known as a browser, such as Netscape's Navigator or Microsoft's Internet Explorer. The browser and web server communicate using the hypertext transfer protocol.

WWW pages can include a wide variety of data types including text, images, animations, audio or video. The web server specifies the type of data using a MIME content type. MIME stands for multipurpose Internet mail extension and was originally designed for sending multimedia e-mail. The browser can handle some MIME content types, for example text/plain, text/HTML or image/GIF, but other formats such as application/postscript, audio/MPEG or video/MPEG are normally handled by a separate application known as a helper application or a plug-in.

A Web page is identified by a uniform resource locator (URL). HTTP is not the only protocol supported by browsers (e.g. FTP and Telnet are also supported), so a URL both specifies the file to be retrieved and the protocol to be used. In general a URL takes the form:

<protocol> :// <hostname> : <port> <directory> <filename>

For example <http://www.bt.com> is the URL of BT's home page. This specifies HTTP as the protocol to be used and www.bt.com as the hostname — the name of the web server where the page is stored. The port, directory and filename default to port 80, the root directory and the filename 'index.html'. The URL is actually <http://www.bt.com:80/index.html>.

The components of an HTML document may be distributed across a number of servers — only the HTML framework has to live on the web server. The base document simply includes a URL for each component and an

instruction to the WWW client to automatically fetch that component.

The content presented at a WWW client may simply be stored files or may be generated dynamically through the execution of a program on the server. The standard means of communication between the web server and other programs is the common gateway interface (CGI). A common usage of CGIs is the translation of an HTTP request into a query of a database on the server, e.g. to perform a search on the information stored.

4. Structure of the net

As the Internet has grown, existing networks have been connected in an *ad hoc* fashion. The result (as illustrated by a piece of the Internet in Fig 2) is a chaotic mesh of networks owned by different enterprises.

Typically when a packet traverses the Internet it will enter and leave the Internet via Internet access providers and may pass through any number of backbone providers on its route. Customers may connect to their access provider's point of presence (POP) via dial-up lines (PSTN or ISDN) or leased lines (as shown in Fig 3).

Internet access providers may themselves be backbone providers. Backbone providers are connected to other backbone providers either via peer agreements (where they agree to carry each other's traffic) or via an exchange such as LINX or CIX (depicted in Fig 2 as circles). Alternatively an access provider may be a re-seller which has bought a connection from a single backbone provider. The chaotic nature of these commercial agreements means that packets may be routed via non-optimal routes. For example, a lack of peer agreements can result in different Internet access

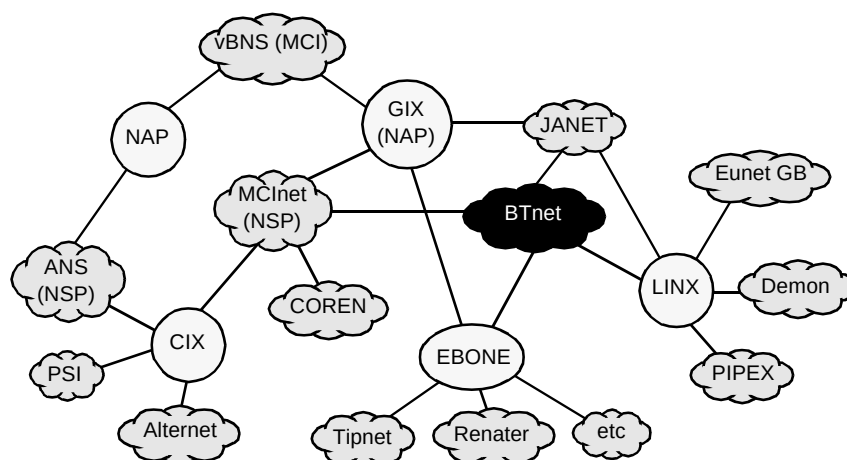


Fig 2 A piece of the Internet.

providers within the same country routing to each other via the US.

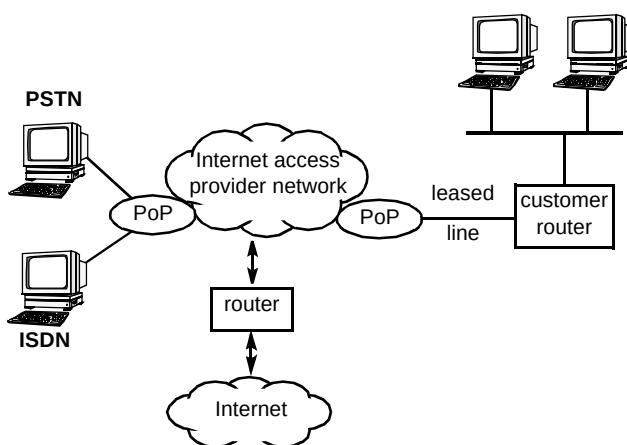


Fig 3 Accessing the Internet.

Commercial pressure, however, is leading to a more hierarchical structure using national Internet exchanges and the emergence of global backbone providers such as BT and MCI's Concert InternetPlus. The current Concert InternetPlus network is illustrated in Fig 4.

5. Intranets and firewalls

Internet technologies also offer major benefits when deployed on an internal company network, or intranet. Indeed, a multi-billion pound market for intranet platforms

is predicted over the next few years. There have been several definitions of an intranet, but basically, an intranet is an internal company network based on the Internet protocol, and which makes use of other Internet technologies such as WWW server and client technology, search engines, e-mail, etc. An intranet may also include links that cross the public Internet to connect together a company's different local area networks. The key benefits and opportunities of an Intranet include the following [7]:

- better communications, both internally and with key business partners — e-mail, mailing lists, bulletin boards, internal 'usenet' type newsgroups, audio-conferencing, videoconferencing, and audio and video broadcasting,
- more effective publishing and distribution of information within an organisation — most companies have a number of internal publications (employee handbooks, quality management systems, company reports, etc) which have the attendant problems of version control and cost of printing and distribution; intranets solve these problems by having a copy of the documents available on a server.

BT now has 30 000 users of its intranet (WWW and e-mail), with a further 30 000 e-mail users. E-mail and WWW usage within BT are growing at 25% and 100% per month respectively. Estimated annual cost savings are currently in excess of £300M.

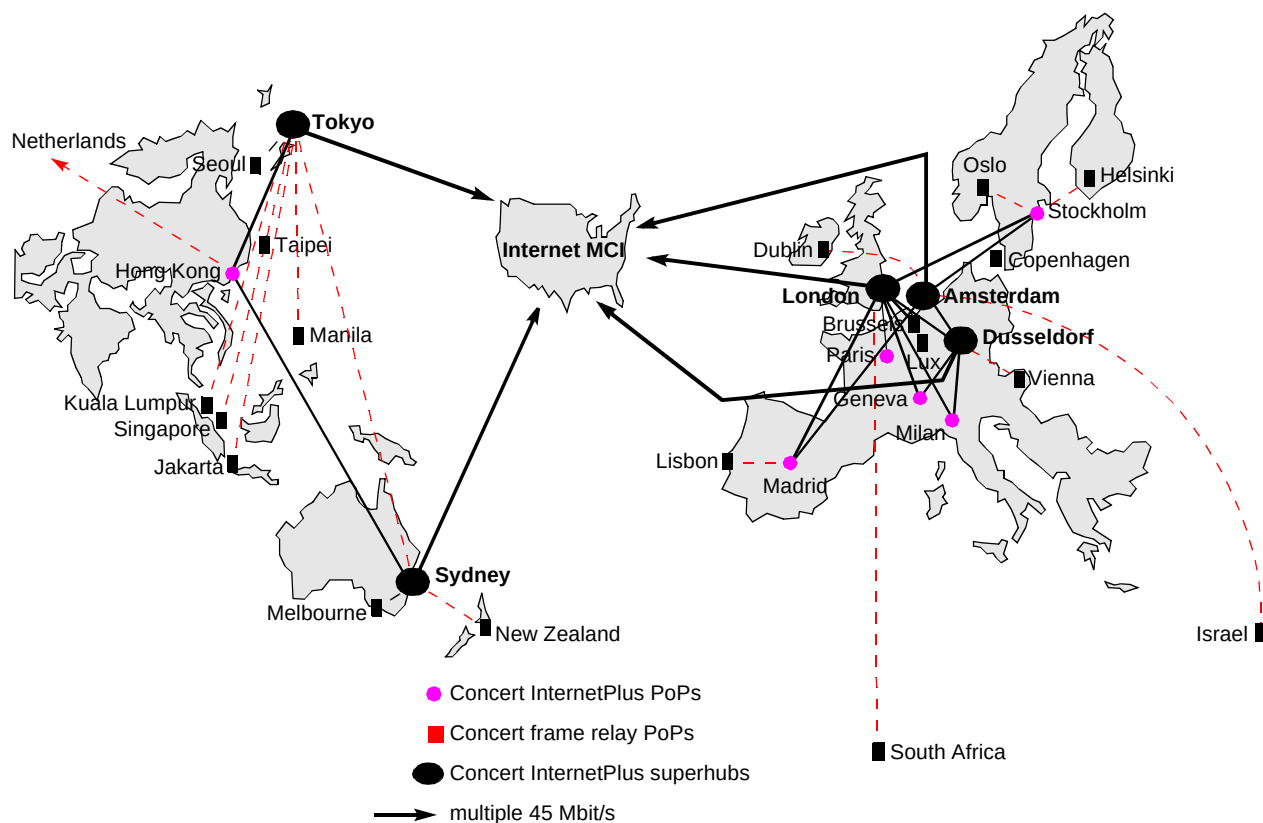


Fig 4 Concert InternetPlus.

Businesses with internal data networks may have a number of reasons for wishing to connect them to the Internet, including:

- marketing and sales,
- connection of their internal network to that of collaborating businesses,
- provision of remote access for employees,
- giving employees access to the global messaging and information services offered by the Internet.

However, connection to the Internet immediately raises concerns over security and a number of potential risks, including access by hackers or competitors to sensitive company information, or the interception of messages that pass over the open Internet. The latter concern can be addressed by encryption, whereas control of access to and from the company network can be achieved by installing a 'firewall' at the interface between the internal network and the Internet (see Fig 5).

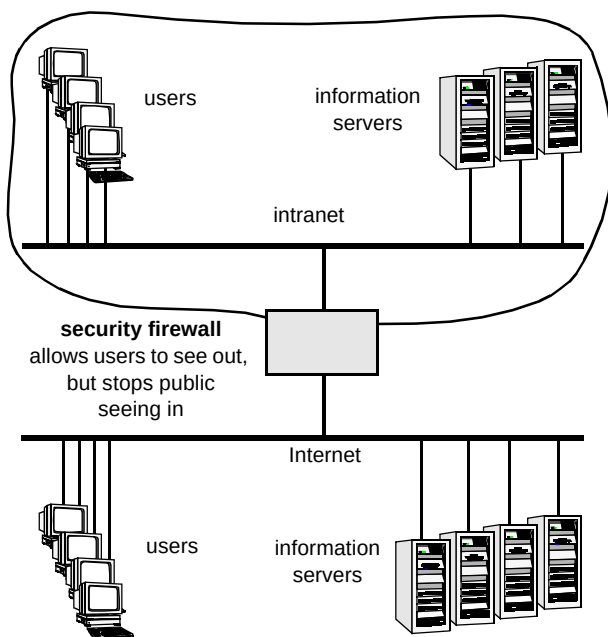


Fig 5 Security firewall between an intranet and the Internet.

Firewalls are installed either on a router or computer at the network interface and fall into two main types, namely packet screens or proxy servers. Packet screens examine each packet passing to and from the internal network, for example to identify the source or destination IP address. Packets are either allowed through or discarded depending on a set of rules determined by the firewall security policy. Proxy servers work on a different principle and control the type of services which are allowed to be accessed by via the firewall, for example WWW or e-mail. Proxy servers can be extremely secure and can also support strong user

authentication. However, they are not very flexible and present proxies do not support new services such as video or multicast. Many variants on these simple principles can be implemented and a fuller description of firewall technologies is given in Hubbard and Sager [8].

6. Electronic commerce

6.1 Trading on the Internet

Current sales of goods over the Internet are modest by comparison with traditional shopping or even mail order. Yet, annual sales over the Net are forecast to rise from their current levels of around \$400M — \$500M to in the order of \$10Bn by the year 2000 [9]. Why is it that such massive growth is predicted? It is not only the large and growing number of Internet users, but also the potential of a multimedia WWW as an environment for the display and marketing of goods, the potential of search engines or software agents to match buyers and sellers around the world, and the emergence of a wide range of electronic payment and banking services. Large-scale electronic commerce is currently being held back by a lack of trust in the security of on-line payment systems, and also by a lack of sophistication in search tools and market-matching services. However, the rapid development of secure payment technology and the involvement of trusted companies in supporting electronic commerce will inevitably overcome these barriers. When they will do so is still open to some speculation, but a steadily accelerating growth in electronic commerce is a more likely scenario than a sudden step change.

Many major shops or companies now have on-line shops through which goods can be ordered, and there are also on-line malls which host hundreds, and in some cases thousands of shops. More powerful electronic market applications are also emerging which offer businesses the potential of large cost savings over traditional methods of finding trading partners. These include buyer-led applications which allow buyers to enter requests for quotations which are then forwarded to appropriate sellers to respond to. An early example of a market matching service is Auto-by-Tel [10] through which consumers wishing to buy or lease a new car can enter their requirements through the World Wide Web. In early 1997, 40 000 consumers per month were submitting requests for quotations to Auto-by-Tel's 1500 strong dealer network. The service is free to consumers, with registered dealers paying a subscription fee. Other services are aimed at the business-to-business market, for example TRADEx [11]. Originally developed for the wholesale computer product market, it has now developed into being a general market-matching service in which buyers can search for, and purchase, products from a catalogue of products and services from a number of sellers.

6.2 Cryptography

Cryptography provides the security necessary for both privacy in communications [12], and also for payments to be made over the Internet. Two forms of encryption are in common use, namely symmetric encryption and asymmetric encryption (the latter is also known as public-key cryptography). Both use cryptographic algorithms and keys to encode and decode data, where keys are parameters used in the mathematical encryption process.

In symmetric key systems, the sender and receiver must use identical keys. The security of symmetric key systems depends on protecting the secrecy of the keys, therefore key distribution requires a separate, secure distribution channel — a disadvantage as far as widespread use in electronic commerce is concerned. However, symmetric encryption algorithms are much faster than public-key algorithms and are therefore preferred for the encryption of large data files. The Digital Encryption Standard (DES) is very well known and is widely used for symmetric encryption systems used for communications.

Public-key schemes use pairs of related keys. Each user has a private key, which is kept secret, and a public key which is published and readily available to others. Public key encryption will enable a wide range of 'trust' services required in electronic commerce [13], in addition to its use in the emerging secure electronic transaction (SET) standards. Encryption of a message with the recipient's public key provides confidentiality — the owner of the corresponding private key is the only person who can read it. This approach is often used as a secure way of exchanging symmetric keys. Encryption of a message with the sender's private key provides both authentication and integrity — if the message can be decrypted correctly using the sender's public key, the claimed identity of the sender and the integrity of the message can be verified. This process, combined with the use of one-way hash functions, is the basis of digital signatures, which can be appended to a document to provide a powerful replacement for a hand-written signature.

The overall security of key-based systems is of course critically dependent on the strength of the cryptographic algorithms, and also on the validity and security of the keys themselves. The strength of the encryption increases exponentially with key length. Unfortunately, there is a conflict of interests between the user's desire for the security of his data, and the concern of governments that secure encryption can support illegal or terrorist activities. For this reason, the US government limits the strength of cryptography that can be exported unless part of the key is 'escrowed' to allow the authorities to have back-door access. Some relaxation of this position has recently taken place, but the debate continues. Smart cards provide a good solution for secure storage of private keys and they will also

support digital signature generation and provide enhanced user authentication. As far as public keys are concerned, users need to trust that these keys are valid, and have not been substituted. This guarantee can be provided by a certification authority which issues a certificate to bind the identity of a user to his or her public key. The certification authority digitally signs the certificate using its private key to guarantee the integrity of its contents.

6.3 Payment schemes

The most widely available method of making a payment on the Internet is currently by credit card. Despite the perceived security risk (which is after all no greater than giving your credit card number to someone over the phone), the credit card payment mechanism offers the user several advantages including the protection of the law in the case of fraudulent use, and the assurance that the merchant must have a relationship with a bank in order to accept card payments. Merchant sites that communicate directly with banks are relatively complex and expensive to implement. To reduce costs and allow smaller merchants to implement on-line commerce, merchants can use a transaction switch, or payment gateway, which provides centralised credit card authorisation and settlement. With both methods network level SSL (secure sockets layer) encryption is used to encrypt credit card details that are passed over the network. Outside the US key lengths for SSL are currently limited by US export laws, but despite this limitation SSL is accepted by financial acquirers. Transactions based on SSL require the merchant to run a secure webserver with a certificate from an approved certification authority.

Major industry players including Mastercard and Visa are working together to develop SET (secure electronic transactions) [14] as a standard for all credit card payments over the Internet. The purchaser's credit card information is encrypted using the public key of the card acquirer, and the merchant (who is unable to read the credit card details) adds his merchant identifier and signs the message before passing it on to the acquirer for confirmation of the payment (Fig 6). Based on public-key cryptography, SET also provides software to collect, exchange and verify certificates.

Whereas SET is now emerging as the likely standard for credit card-based transactions, in the field of electronic cash, or e-cash, there are a number of solutions and no clear standard has emerged.

Mondex [15] provides electronic cash stored within a chip on a card and has the added bonus that it can be used in the real world to buy items in shops (as in the recent trial in

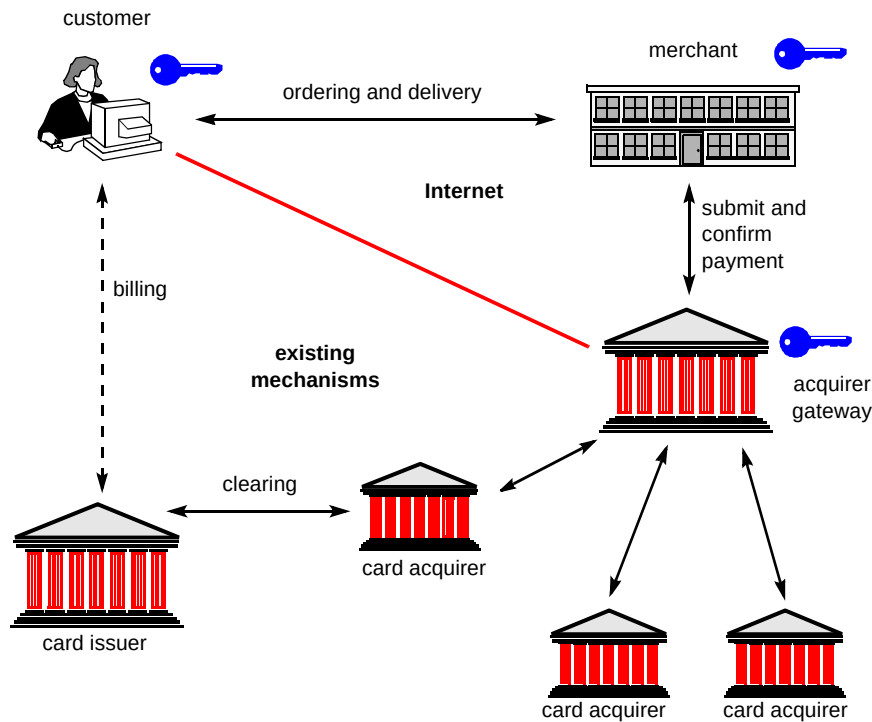


Fig 6 Secure credit card payment model.

Swindon in the UK), for making person-to-person transactions, or over the Internet. Software-based wallet or cash systems that are being offered include Cybercash [16] and Digicash [17]. The Digicash system provides users with digital coins which are signed by a central banking authority using public-key cryptography and are stored by users on their local computer. The user can spend this digital money at any shop on the Internet which accepts e-cash, without the need to open an account there first, or having to transmit credit card numbers. Person-to-person payments can also be performed with e-cash. The strengths of the system are its security and the anonymity of a cash system, but scalability and performance may be problematic because the bank is involved in all transactions, and a database of spent coins is required.

Because payments below a few pounds using a credit card or e-cash over the Internet are likely to be uneconomic due to the transaction overhead, a number of micropayment systems are being developed for the purchase of low-value items or information. Examples of such systems are Payword, Millicent [18], MicroMint and QuickPay [19]. Payment systems can be much cheaper to implement if audit trails are not required for each transaction and efficient protocols or algorithms can be used. Indeed protocols must be efficient if large volumes of low-value transactions are to be processed. Both pre-paid and post-paid systems are being developed — the former being analogous to electronic phone cards. QuickPay differs from other payment systems in that it is based, not on cryptography, but on random numbers. Users purchase a

wallet of tokens in the form of random numbers, which will only be accepted by the merchant if presented in the correct sequence. This lack of reliance on encryption makes the technique unaffected by the export regulations which affect some other payment schemes. A fuller description of payment systems is given in Putland et al [19].

7. Three-dimensional Web

Only a few years after the WWW was first created, 3-D browsers are now making possible a whole new range of applications. 3-D models of objects can be built into Web pages and viewed from all angles. Very complex models can be built, for example, to enhance on-line shopping catalogues, to display engineering designs, to model complex data sets, or to model the layout of a new housing development. Perhaps the most exciting development is the ability to create shared three-dimensional worlds in which communities of people can meet. Users in the 3-D world are represented by avatars (3-D characters) which can move around in the world and interact with the other visitors. Drawing on the news-group experience, these 3-D worlds can be designed to support groups of users with a common interest or hobby. Most shared worlds on the Web support only typed text messages for users to communicate, limiting their appeal. However, the developing capability of the Internet to support sound, and particularly spatially resolved sound within these 3-D worlds, will open up the potential for mass-market applications. The avatars representing visitors to these virtual worlds are often abstract or amusing in form. However, humanoid models, with realistic facial

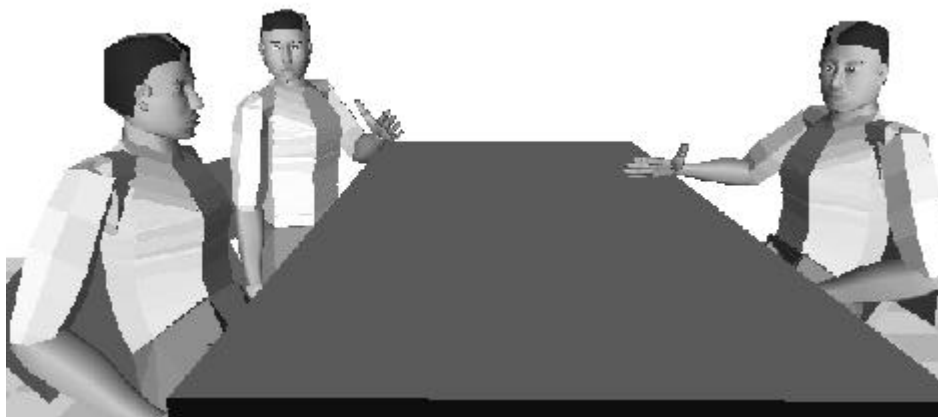


Fig 7 A 3-D virtual conference (from Machin and Shepherd [20]).

features and voice synchronised lip movements offer the potential for conferencing applications which overcome some of the limitations of large audio conferences (see Fig 7) [20]. The person who wishes to speak next can easily attract the attention of the chairman in a virtual conference room.

Future conferencing applications with enhancements such as facial recognition, combined with humanoid avatars with realistic faces, may prove to be more intuitive to use than videoconferencing with multiple split screen images of the participants.

The technology underpinning 3-D applications on the WWW is the virtual reality modelling language, or VRML. Although spawned by the huge success of the WWW, VRML is not an extension of HTML. The needs of a 3-D modelling language are quite different. VRML1 was first published in 1995 and provided the ability to define 3-D objects or worlds, hyperlinks to other 3-D worlds or to standard Web pages, and the ability for users to navigate around.

Later versions of VRML allow objects to interact with one another and support video and spatially resolved sound within worlds. Compound animations such as a walking human body can also be created. The performance of 3-D applications, in terms of download time and speed of response or movement within worlds, is dependent on the complexity of the models, but with careful design, very acceptable performance can be achieved on today's top-end PCs. The inexorable rise in PC performance and memory capacity, together with 3-D PC cards, will overcome performance issues.

There are a number of other 3-D Internet developments which are not based on the open VRML standard, most notably Superscape — a powerful authoring and viewing environment.

8. Network computing

Network computing is a new approach to computing in which all software and data is stored remotely on servers in the network and downloaded to the user's client as and when required [21]. Unlike mainframe computing the application is executed locally. Unlike PC computing, the application is stored in the network and not on the local machine. The main advantages of network computing are reduced version management, and rapid access to new applications and services.

8.1 Java

The foundation of network computing is the programming language Java. What distinguishes Java from other competing languages is the fact that the Java virtual machine (the software required to execute Java) is available within every WWW browser. This means that virtually every networked computer is, or will be, capable of running a Java program.

Java came to the fore in 1995 as a technology to activate the Web. At a time when WWW pages tended to be a static combination of text and graphics, Java offered animation and the full power of a computer programming language in the browser.

To run a Java program it must first be compiled into an intermediate form known as a byte code. The byte code can then be interpreted by the Java virtual machine into host specific instructions. It is this interpreted nature of Java that allows a single version of a Java program to run 'anywhere'.

Once compiled, a Java program is typically made available on a networked server and advertised via WWW pages. To execute the program, the user simply clicks the appropriate hyperlink and the software is automatically downloaded and executed on the user's machine. Thus the

user has a simple means of accessing software, does not need to worry about upgrading old software, and new services can be invoked without the need to first install the required software. Increasingly Java applications can expect to be seen running independently of the browser.

8.2 Issues

Its portability and pre-eminent position on the World Wide Web has assured Java an important role in the development of the Internet. However the extent of its success depends on two main issues.

- **Bandwidth** — downloading software and data every time a program is launched can be painfully slow if the available bandwidth is poor, so adequate bandwidth is clearly required. Caching techniques such as Marimba's Castanet [22] (which allow software to be stored locally and only requires it to be downloaded if the server copy has been changed) will also help.
- **Security** — to protect against malicious software and attack from fraudsters a number of measures are required. These include protecting the integrity of remotely stored software and data, and, in the case of accessing another party's software, assuring the identity of the party and its ownership of the software.

9. Real-time services and the integrated services architecture

Until recently workstation performance and insufficient bandwidth have been critical factors in stifling the development of real-time services. However, with exponential improvements in workstation performance, the standard inclusion of multimedia capabilities in PCs, and advances in compression techniques, the average owner of a new PC is now capable of exploiting real-time network-based services.

This has led to an explosion in the development of products delivering real-time services over the Internet. These include Internet telephones, audio and video streaming applications, videoconferencing, whiteboards, shared applications and shared virtual worlds. The prospect of 'free' calls has led some enthusiasts of Internet telephony to proclaim the end of the public switched telephone network (PSTN). In fact the fundamental costs of Internet telephony and PSTN telephony are similar and Internet telephony has a long way to go to match the convenience, ease of use, call quality and customer service of PSTN telephony. Nevertheless Internet telephony has one major advantage over PSTN telephony — flexibility. With compression in the end-system it is able to quickly take advantage of latest compression algorithms.

Unrestricted by fixed 64 kbit/s channels, Internet telephony can, in principle, always select the optimum balance between cost and quality. Furthermore Internet telephony is easily integrated with other media or other computer applications. For these reasons today's Internet real-time applications signal a fundamental change in the delivery of future real-time services. Consequently there is much interest in providing a gateway for telephone calls between IP networks and the PSTN as depicted in Fig 8. Banthorpe et al discuss further the technical and economic issues of Internet telephony [23].

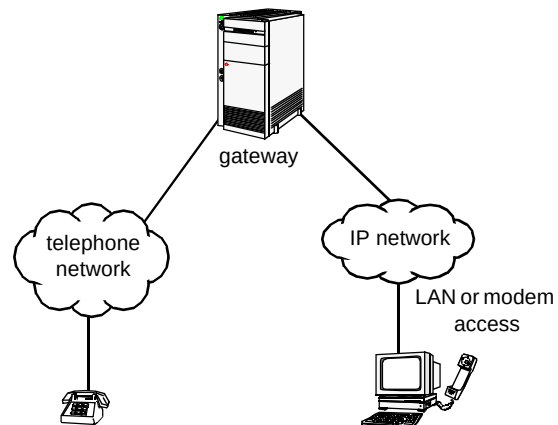


Fig 8 A gateway for Internet telephone calls between IP networks and the PSTN.

Today's real-time applications on the Internet are currently offered on a 'best effort' basis and are vulnerable to the effects of network congestion, such as packet loss, increased delay and wide variations in network delay. Consequently their widespread deployment depends on adequate dimensioning of the network or on the development and deployment of mechanisms to control the quality of service in the network. To this end the IETF is developing the integrated services architecture (ISA) — a framework that aims to offer control over the quality of service provided by a network. ISA allows applications to prioritise their traffic and to request network resources to implement those priorities. Two major components of ISA are IP multicast and resource reservation protocol (RSVP). IP multicast supports the efficient distribution of flows of information to many receivers. It works by building a multicast tree such as that shown in Fig 9. Because information sent to all receivers only traverses each branch of the tree once, it is extremely efficient and highly suited to the of high-bandwidth media such as audio and video. RSVP allows receivers to place reservations for their part of the multicast flow. Because it is receiver driven, different receivers can make reservations of different qualities. For example, some receivers might reserve resources for audio

only, some for black and white video and some for full colour. A full discussion of ISA is provided by O'Neill [24].

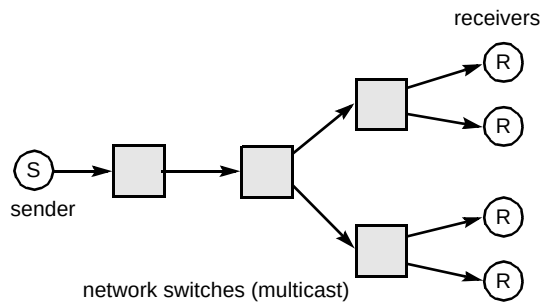


Fig 9 Network multicast.

The quality of service delivered to the user depends on the quality of service delivered by the network and the operating systems, as well as on the application protocols. Very often the quality of service requirements for real-time applications are in conflict and trade-offs are required. For example retransmissions used for reliable delivery will lead to increased end-to-end delay and increased jitter (variation in end-to-end delay). Furthermore, it becomes increasingly difficult to guarantee delivery to all parties as the number of parties increases. Ideally it should be the application that has control over the various trade-offs that need to be made. These issues are discussed as part of a general introduction to real-time applications in Rudkin et al [25]. Based on these principles the paper outlines an application architecture for real-time services (with particular reference to audiovisual services). The enhancement of the Internet and intranets to support real-time services is one of the most exciting developments under way. It promises a future in which a single corporate network can be used for voice, data and all other media. It promises excellent computer/telephony integration (CTI) and it promises interactive entertainment such as shared virtual worlds and inhabited TV. The determining factors in the growth of these services are the deployment of the ISA and the roll-out of high-bandwidth access networks, the development of application interfaces that the mass market finds easy to use, and time for consumers to become confident with the security measures being introduced.

10. Conclusions

This paper has charted the development of the Internet from its military and academic origins to its current status as a global network for multimedia information publishing and retrieval. The key to the Internet's initial growth was the capabilities of the IP suite — its resilience against failures and its ability to enable the interconnection of many different networks. The continued growth and evolution of the Internet is the result of its modular, loosely coupled architecture which has enabled the rapid introduction of new protocols which can support new services. The poten-

tial of the Internet to support electronic commerce and real-time services has already been demonstrated, albeit on a relatively small scale. The benefits of using Internet technologies in internal corporate networks (intranets) are very clear. The next stage will see the widespread development of electronic commerce services and dramatic growth in intranets. Real-time services are likely to make an impact first on IP-based local area networks in the form of applications such as desktop conferencing, or video-based training or news services. The future will see real-time services with quality of service guarantees across the global Internet.

There are several issues still to be overcome:

- perceptions that the Internet is insecure (if its full potential for electronic commerce is to be realised), concerns over the strength of encryption products, and a range of acceptable payment and trust services,
- as more recent uses of the Internet, such as Web browsing, Internet telephony, audio and video streaming and conferencing, generate much more traffic than older services such as e-mail or file transfer, charging may be required to balance available bandwidth in the Internet backbone with the explosive demand, and also to fund the provision of significantly more capacity in the future — higher bandwidth access for residential users will also be required to support video-based services,
- further development of the IP protocol suite and of high performance routers and switches will be required to move ahead from the current 'best-efforts' service to an integrated services platform with quality of service guarantees,
- a massive amount of information has been made available via the WWW and current search engines go some way to helping users find the information they are looking for, but further significant improvements are required through the use of technologies such as intelligent agents and natural-language processing — more generally, ease of use will be a critical factor in the mass-market take-up of Internet services.

The Internet will have a major impact on telecommunications operators. Network evolution plans will have to be re-evaluated to encompass the growing capabilities of the Internet protocol suite [24]. Some have claimed that the advent of Internet telephony will undermine the core business of telecommunications operators. We believe that overall, the opposite effect will prevail, and that the Internet will stimulate massive growth in world-wide communications. In addition to the growth in total 'bit' transport, the Internet will also stimulate a host of new business opportunities including Internet service provision for business and residential customers, Web hosting, intranets, on-line services, and electronic commerce. Telcos

have been struggling to find the 'killer application' that will justify the massive cost of rolling out broadband access to every home. The rapid development of WWW-based services is already leading to massive pressure for more bandwidth in the Internet backbone. It seems inevitable that a WWW which supports multimedia information retrieval, commerce applications, conferencing, 3-D worlds and streamed audio and video services will raise customer expectations and make the provision of high-bandwidth access a necessity, not an option.

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